UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

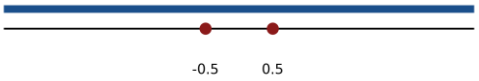
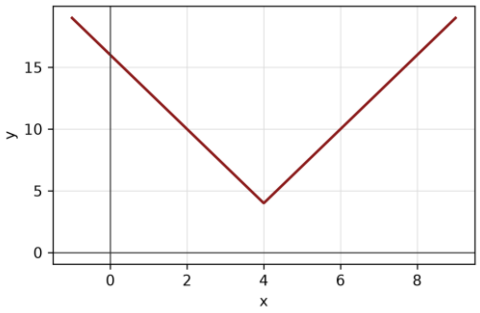
INSTRUCTIONS TO MARKERS

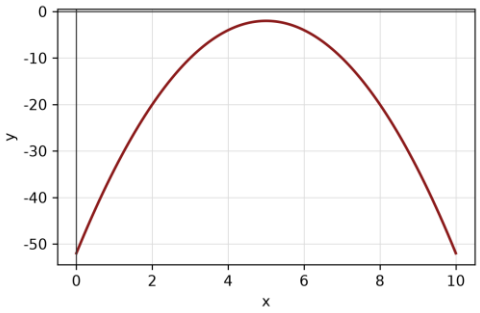
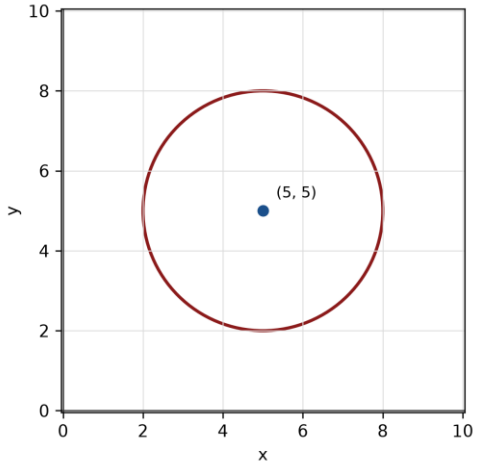
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

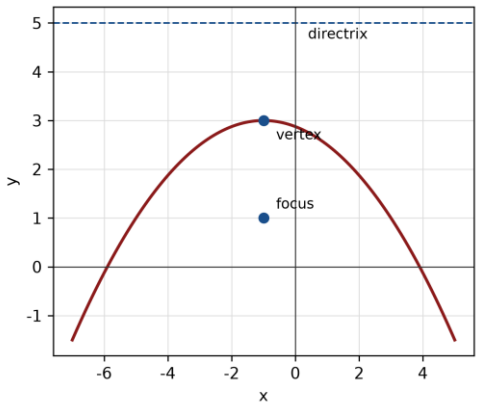
FOR MARKERS' USE ONLY

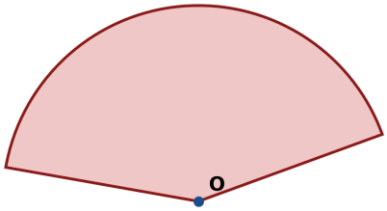
This marking scheme consists of 5 printed pages

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NO	SOLUTION		MARK
1 a)	$\frac{-2x-3}{2x+1}+2\leq 0$	M1 A1	5 m
	$\frac{2x-1}{2x+1}\leq 0$	A1	
	$\left(-\frac{1}{2},\frac{1}{2}\right]$	A1	
			
1 b)	$\lvert 2x+2\rvert-6>0$	M1	5 m
	$-2x-2>6$ or $-2x-2<=-6$	A1	
	$(-\infty,-4)\cup(2,\infty)$	A1	
			
1 c)	$y=3\lvert x-4\rvert+4$	B1	5 m
	vertex $(4,4)$	A1	
	no real intercepts	A1	
			

NO	SOLUTION		MARK
2 a)	$y = -2(x-5)^2 - 2$ vertex $(5, -2)$ $y(0) = -52$ 	B1 B1 A1 A1	5 m
2 b)	$f(x) = x^4 - x^2 + 5$ $f(-x) = x^4 - x^2 + 5$ $-f(x) = -x^4 + x^2 - 5$ $f(-x) = f(x)$	M1 A1 A1	4 m
2 c)	$(x-5)^2 + (y-5)^2 = 9$ centre $(5, 5)$, $r = 3$ 	M1 A1 A1	4 m
3 a)	$(-3 + 3i) + (-6 - 4i)$ $-9 - i$	M1 A1 A1	4 m

NO	SOLUTION		MARK
3 b)	$\lvert z\rvert = \sqrt{-7^2 + 24^2} = 25$ $\tan\theta = \frac{24}{-7}, \quad \theta = 106.3^\circ$	M1 A1 M1 A1	5 m
3 c)	$-2x^3 + 10x^2 - 4x - 16$ $f(2) = 0, \quad f(-1) = 0, \quad f(4) = 0$ $-2(x-4)(x-2)(x+1)$	M1 M1 A1	5 m
4 a)	$2x - 2y - z = 9$ $2x + 2y - z = -7$ $2x + y + 2z = 6$ $x = 2, \quad y = -4, \quad z = 3$	M1 A1 A1	5 m
4 b)	$-3x - 8 = x^2 - 6$ $x^2 + 3x + 2 = 0$ $(x+1)(x+2) = 0$ $x = -1$ or $x = -2$ $(-1, -5), \quad (-2, -2)$	M1 A1 A1	5 m
4 c)	$y = -\frac{x^2}{8} - \frac{x}{4} + \frac{23}{8}$ $y = 3 - \frac{(x+1)^2}{8}$ $(x+1)^2 = 24 - 8y$ 	M1 A1 A1 A1	5 m

NO	SOLUTION	MARK
5 a)	$\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$ **M1** $\sin^2(x) = 1-\cos^2(x)$ **M1** $\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$ $\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$ **A1**	5 m
5 b)	$\theta = \frac{5\pi}{6}$ **M1 M1** $A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2\left(\frac{5\pi}{6}\right) = 15\pi$ **A1** 	4 m
5 c)	$\text{Area} = \frac{1}{2}ab\sin C$ **M1 M1** $\text{Area} = \frac{1}{2}(7)(12)\sin 30^\circ = 21$ **A1**	4 m